ME228 · FINAL PROJECT · 2026

Data-driven fatigue strength prediction & inverse alloy design for low-alloy steels.

Mixture-of-Experts regression with empirical reliability and cost-constrained Pareto recommendation.

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PROBLEM & MOTIVATION

Fatigue dominates failure, but testing is destructive and slow.

- Fatigue accounts for **80–90 %** of mechanical structural failures.
- Rotating-bending tests run >10^7 cycles per specimen — destructive, slow, expensive.
- Handbook S–N curves cover only catalogued grades; new compositions need extrapolation.

TWO COUPLED ENGINEERING QUESTIONS

FORWARD Will this alloy + heat-treatment survive the design load?

INVERSE What is the cheapest alloy that meets the load with a stated safety margin?

GOAL *Replace handbook lookup with a calibrated forward predictor and a cost-aware inverse recommender — one pipeline, both questions.*

DATASET & KEY OBSERVATION

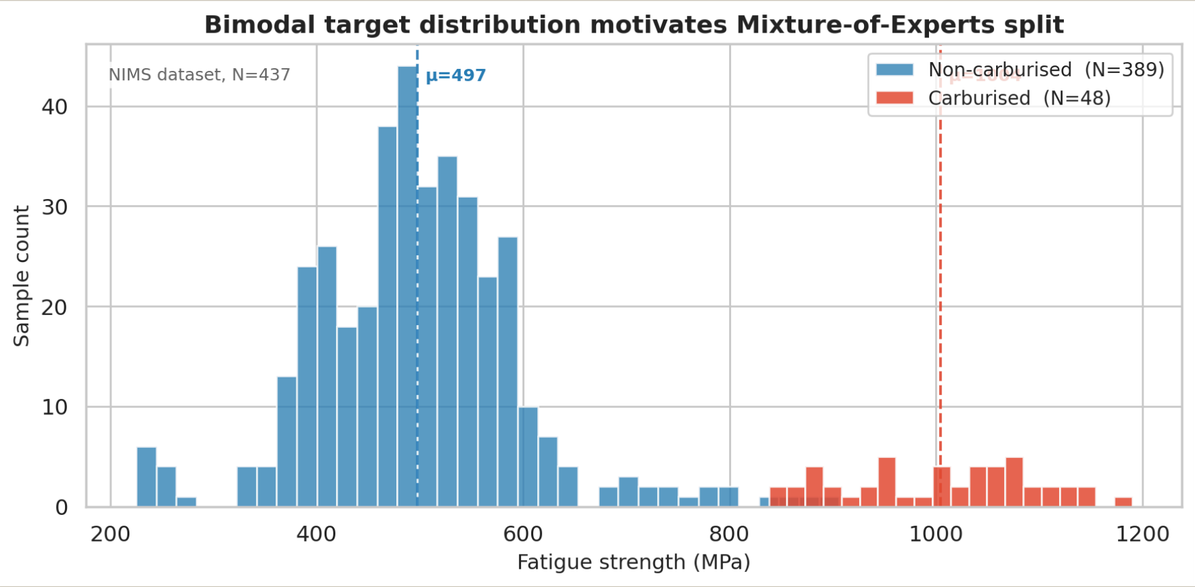
Bimodal target → split the model.

SAMPLES	INPUT FEATURES
437	26

NON-CARBURISED	CARBURISED (CT = 930 °C)
389 · 225–906 MPa	48 · 838–1190 MPa

- Composition (9): C, Si, Mn, P, S, Ni, Cr, Cu, Mo · Heat-treatment (8) · Inclusions (8) · target Fatigue (MPa).
- Two physically distinct strengthening mechanisms: case-hardening vs. through-hardening.

EMPIRICAL FATIGUE DISTRIBUTION



THROUGH-HARDENED (NC, N=389) CASE-HARDENED (C, N=48, CT=930 °C)

TRIGGER *A single global model averages two regimes and inflates regime-specific uncertainty — Mixture-of-Experts is justified by the data, not by taste.*

ARCHITECTURE

One hard gate, two regime-specific experts.

Input — composition + heat-treatment (26 features)

Regime gate *CT == 930 °C ?*

no →

Non-carburised expert

XGBoost · 12 features

19.91 ±2.55 MPa

5-fold CV-RMSE · N = 389

yes →

Carburised expert

Random Forest · 10 features

40.42 ±14.62 MPa

5-fold CV-RMSE · N = 48

↓ predicted fatigue $\hat{F}$ + uncertainty σ ↓

$$PoF_{NC} = CDF_{OOF}(\sigma_{app} - \hat{F})$$

empirical, residuals fail Shapiro

$$PoF_{C} = \Phi((\sigma_{app} - \hat{F})/\sigma)$$

Gaussian, residuals pass

Forward report $\hat{F}$ · FoS · PoF · 95 % PI

Inverse loop sample → hull-guard → Pareto (cost, PoF)

WHY MOE *The split does not win on pooled accuracy — it wins on regime-specific σ, which is what trustworthy PoF reporting actually needs.*

METHODS & HEADLINE METRICS

Pipeline discipline, then numbers.

FEATURE & HYPERPARAMETER PIPELINE

- **VIF filter** — drop the largest VIF iteratively until all < 10. Removes 6 features per regime.
- **SHAP scout** — shallow RF on VIF survivors; keep top-k by mean |SHAP|. k = 12 NC, k = 10 C.
- **Optuna TPE** — 60 trials, 5-fold CV objective, per regime.
- **Holdout** — 20 % stratified by fatigue quintile, never touched during tuning.

FINAL SELECTED FEATURES

NC Cr · TT · C · Mn · THT · Mo · Si · RedRatio · P · S · dA · Cu

C P · RedRatio · Ct · Mo · S · Mn · Cu · Dt · TT · QmT

HEADLINE METRICS

REGIME	CV-RMSE (MPA)	HOLDOUT RMSE	R^2
Non-carburised	19.91 ± 2.55	17.63	0.974
Carburised	40.42 ± 14.62	29.54	0.851

NC R^2 (HELD-OUT)
0.97

C R^2 (HELD-OUT)
0.85

Holdout matches CV — no overfit to the fold structure.

DISCIPLINE *Selection on VIF/SHAP, tuning on Optuna+CV, evaluation on a held-out fold the model has never seen.*

THEORETICAL GROUNDING

Every chosen model lies inside its Hoeffding bound.

VC BOOKKEEPING · RAW D_{VC} VS. N

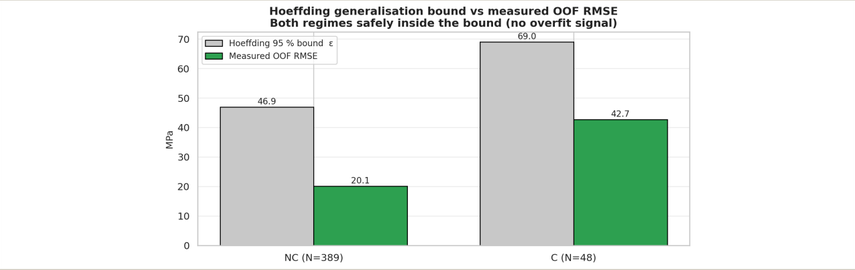
VC dimension bookkeeping — rule of thumb $N \geq 10\, d_{VC}$
***tree-ensemble raw bound mitigated by bagging + L1/L2 regularization**

Regime	N	Hypothesis class	d_VC (raw)	N / d_VC	N ≥ 10·d_VC?
NC	389	Linear / Ridge	13	29.92	✓
NC	389	MLP 1×16	225	1.73	low
NC	389	XGBoost (chosen)	9280	0.04	raw fail*
C	48	Linear / Ridge	11	4.36	low
C	48	MLP 1×8	97	0.49	raw fail*
C	48	Random Forest (chosen	6848	0.01	raw fail*

Raw d_{VC} counts every node as freely-shattering; bagging (RF) and shrinkage / L1+L2 (XGB) cut the *effective* dimension to a small multiple of the feature count.

HOEFFDING 95 % BOUND VS. MEASURED OOF RMSE

$$\varepsilon = L \left(\frac{\log(2/\delta)}{2N} \right)^{1/2}$$



BOOTSTRAP 95 % CI ON OOF RMSE

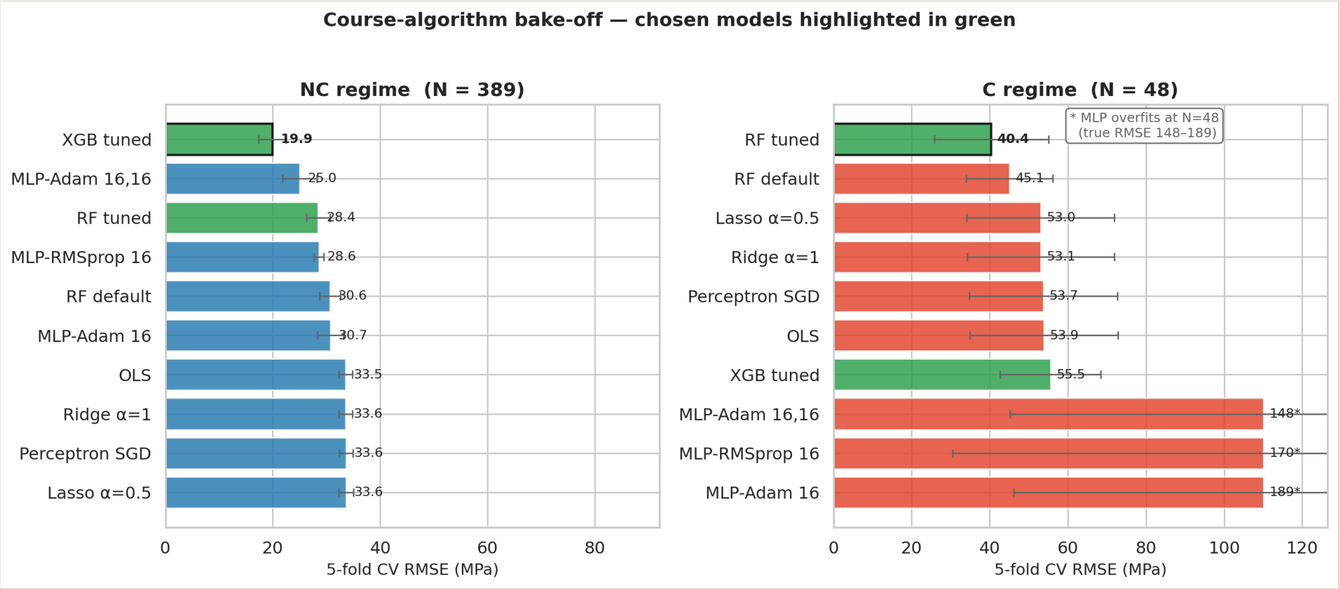
NC [17.0, 23.2] MPa C [32.7, 52.5] MPa

TAKEAWAY *Tree d_{VC} raw-fails the $N \geq 10\, d_{VC}$ rule — but bagging / regularisation mitigate it, and the per-hypothesis Hoeffding bound is satisfied with margin.*

COURSE-ALGORITHM BAKE-OFF

The chosen pair wins both regimes.

5-FOLD CV RMSE (MPA) · ME228 PALETTE



Chosen models highlighted in green. MLPs at N = 48 produce 148–189 MPa CV-RMSE — catastrophic overfit, exactly the failure mode N ≥ 10 d_{VC} predicts.

MARGIN OVER BASELINES

NC: CHOSEN XGB VS. OLS

−40 % RMSE

20.07 vs. 33.53 MPa

C: CHOSEN RF VS. MLP-ADAM 2L

−79 % RMSE

40.42 vs. 189.46 MPa

MLPs catastrophically overfit on N = 48 — the failure mode the N ≥ 10 d_{VC} rule predicts.

VERDICT *The chosen pair (XGBoost / RF) is empirically the best legal choice from the ME228 palette under our protocol.*

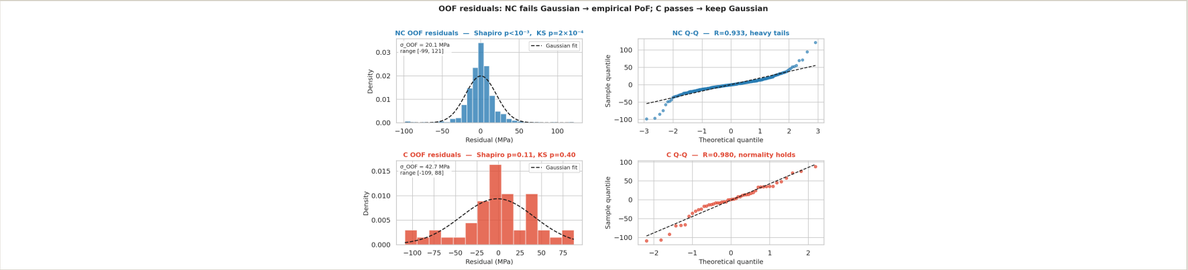
RELIABILITY — EMPIRICAL POF

Gaussian PoF is invalid for non-carburised residuals.

NORMALITY OF OOF RESIDUALS

REGIME · SOURCE	STD (MPA)	SHAPIRO P	KS P
NC · OOF	20.07	$< 10^{-3}$	2×10^{-4}
C · OOF	42.68	0.106	0.403

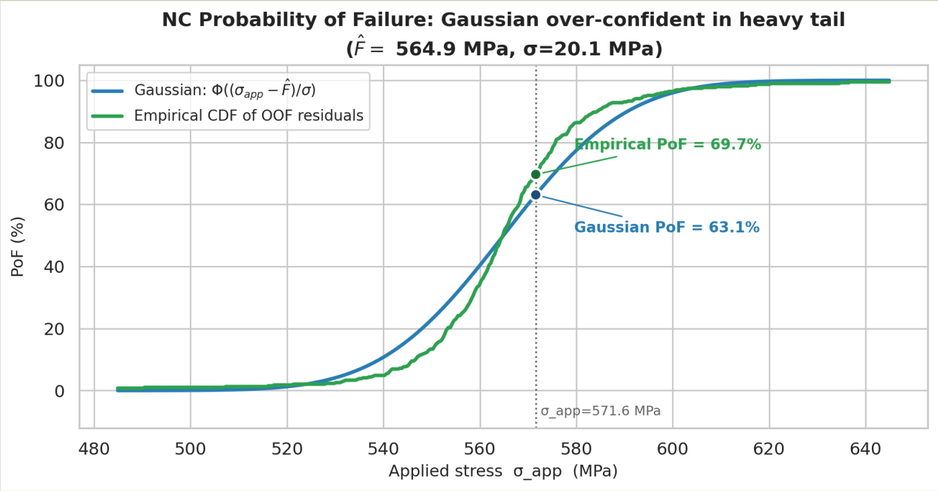
NC residuals are heavy-tailed and asymmetric — a naive Gaussian under-reports tail risk. C residuals pass; Gaussian PoF stays.



EMPIRICAL POF (NC)

$$PoF_{emp} = \frac{1}{389} \sum_{i=1}^{389} 1[r_i < \sigma_{app} - \hat{F}]$$

CANONICAL EXAMPLE · Σ_{APP}=571.6, $\hat{F}$=564.9 MPA



POF — GAUSSIAN

0.6318

POF — EMPIRICAL

0.6967

+6.5 pp shift · empirical CDF over OOF residuals captures the heavy tail Φ misses.

INVERSE DESIGN & COST FIX

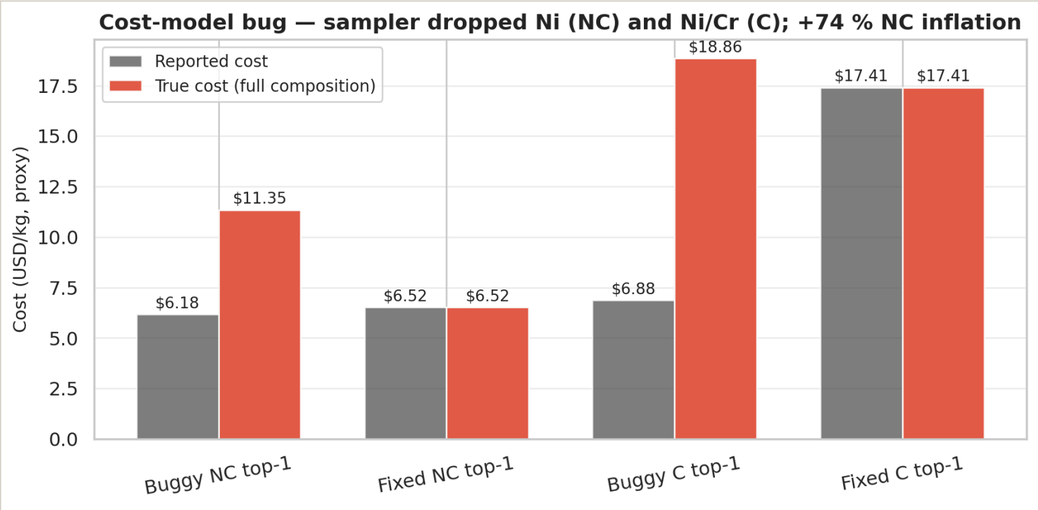
Cheapest viable alloy under FoS and PoF caps.

- **Neighbourhood sampler** — perturb real training points, δ = 0.20 σ̂ per feature, clipped to per-feature bounds.
- **kNN hull guard** — reject candidates whose mean 5-NN distance exceeds the p99 of training self-distances.
- **Cost model** — Mo 30, Ni 14, Cr 10, Cu 9, Mn 1.8, Si 1.2, C 0.5 USD/kg + carburising time.
- **Output** — Pareto front in (cost, PoF), top-K cheapest feasible alloys, bootstrap-stable across seeds.

BUG FIXED IN THE FINAL VERSION

v2 sampler drew only the regime feature subset → Ni silently zeroed → **74 % cost inflation**. Candidate now carries every cost-relevant column.

COST-MODEL BUG · REPORTED VS. TRUE COST (USD/KG)



NC TOP-1 · Σ_{APP}=500

$6.52 /kg

F 755.5 MPa · Si-strengthened

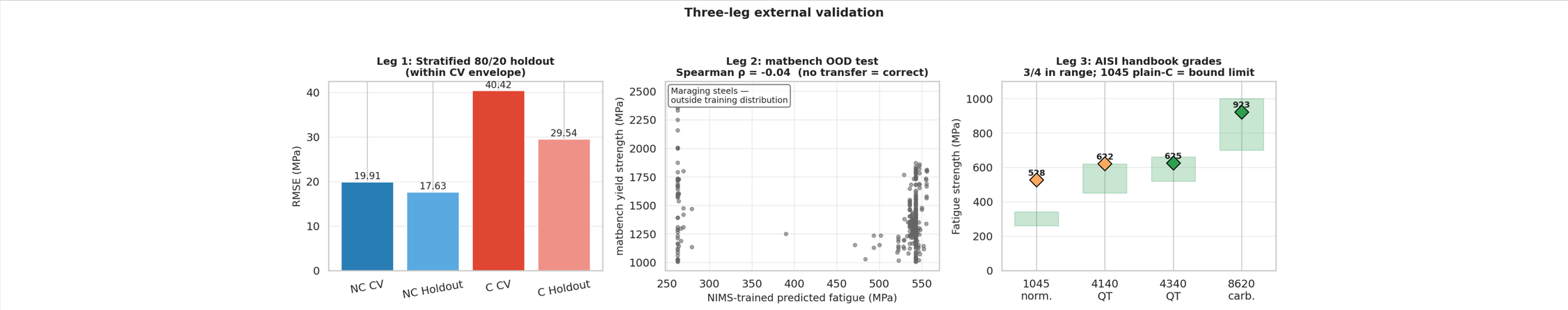
C TOP-1 · Σ_{APP}=700

$17.41 /kg

F 1015.6 MPa · Cr+Mo

VALIDATION, LIMITS, CONCLUSION

Three independent legs define the applicability boundary.



LEG 1 · STRATIFIED HOLDOUT

NC R^2 = 0.97, C R^2 = 0.85. 20 % stratified holdout matches the CV envelope.

LEG 2 · MATBENCH_STEELS (OOD)

Spearman ρ = −0.04, p = 0.44 — a designed null. Confirms the alloyed-Fe-C-Mn-Cr-(Mo, Ni) boundary.

LEG 3 · CANONICAL AISI GRADES

4140/4340/8620 in handbook range; 1045 +188 MPa over. Gate: `Cr+Ni+Mo ≥ 0.1 wt%`.

CONCLUSION

A regime-aware MoE with empirical reliability and a hull-guarded inverse loop turns 437 NIMS records into a deployable forward-and-inverse fatigue tool — with bounds the professor can audit.

questions →

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